

Antonio Vasconcellos Mac Dowell

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Summary

I am a Brazilian-Portuguese graduate Mathematics student from Rio de Janeiro, currently based in Paris, specializing in Poisson geometry, integrable systems and representation theory. Experienced in developing numerical algorithms, risk management tools and teaching.

Education

PhD in mathematics

2024 – Present

- **Institution:** Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio)
- **Advisor:** Graham Smith
- **Scholarship:** Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)
- **Research topic:** Integrable systems, Poisson geometry

Master's degree in Mathematics

2021 – 2024

- **Institution:** Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio)
- **Advisor:** Carlos Tomei
- **Scholarship:** Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)
- **Dissertation:** [Recovery of Jacobi matrices from spectral data](#) 
- **Brief description:** Creation and development of an algorithm for inverse spectral theory of tridiagonal symmetric matrices; Numerical experiments showed superior performance compared to existing techniques.
- **Extra activities:** [IMPA](#) : Lie Algebras, Algebraic Topology, Symplectic and Poisson Geometry.

Bachelor's degree in Mathematics

2018 – 2021

- **Institution:** Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio)
- **Extra activities:** Teaching Assistant for Calculus, Linear Algebra and Differential Equations.
- **Math Olympiads:** XII Competencia Iberoamericana Interuniversitaria de Matemáticas (CIIM-2020)

Experience

Teaching Experience [PUC-Rio](#)

March 2024 - Dec 2024

- Discipline: Modelagem e Programação (CTC4002), Undergraduate course, 2 classes of 40 hours each.
- Description: Pre-calculus with an introduction to algorithmic thinking and Python programming.

Risk Analyst Intern [RJ+ Asset](#)

2023 (summer)

- Member of a team developing algorithms for risk management, automating the computation of Value at Risk (VaR) for portfolio optimization with Python and Excel.

Software Engineer Intern

Jan 2021 – Aug 2022

[Fundação COPPETEC](#), [Federal University of Rio de Janeiro](#) 

- Member of an AI project for the State Public Defender's Office, Rio de Janeiro.
- Management of datasets, development of algorithms and meetings with clients.
- The project made significant contributions to the work of public defenders in Rio de Janeiro.

Skills

- **Programming languages:** Python, SQL;
- **Software/Tools:** LaTeX, Sage, Maple, MatLab, Excel, DataGrip, Visual Studio, GitHub/GitLab;
- **Languages:** Portuguese (Native), English (Fluent), French (Level B1);
- **Other skills:** My professional and teaching experiences provided me with strong team work and public speaking skills.

Seminars

I actively participate in seminars and conferences related to my research interests. Here is a list of some talks I gave and seminars I've attended lately:

- Talk: *Random Matrices, Graphs and a combinatorial proof of Wigner's Semi-Circle Law* [↗](#) (PUC-Rio)
- Conference: *Hyperbolic Manifolds, Their Submanifolds and Fundamental Groups* [↗](#) (IMPA)
- Conference: *Systolic and Diastolic Geometry* [↗](#) (IMPA)
- Conference: *Interactions of Poisson Geometry, Lie Theory and Symmetry* [↗](#) (IST-Lisbon)
- Conference: *X Workshop on Conservative Dynamics and Symplectic Geometry* [↗](#) (IMPA)
- Conference: *35º Colóquio Brasileiro de Matemática* [↗](#) (IMPA)
- Conference: *Carlos Tomei 70 anos* [↗](#) (PUC-Rio)
- Talk: *Introduction to Toda lattices* [↗](#) (IMPA)